

A Framework for Clarifying Questions in Consciousness Research

Abstract

Consciousness research encompasses diverse theoretical approaches that often appear contradictory. We propose that many apparent disagreements reflect conceptual misalignment rather than empirical conflict: researchers address different explanatory targets without making these differences explicit. We present a two-dimensional framework with two key distinctions. First, level of analysis: internal mechanisms that may support or constitute consciousness versus external observable indicators used to infer or detect consciousness. Second, system type: biological organisms versus artificial systems. This creates four analytical quadrants that clarify what different research programs investigate. We demonstrate how major theories occupy distinct quadrants. Global Workspace Theory primarily addresses how conscious content enables external report and behavioral control, while Integrated Information Theory targets internal integration structures. Clinical evidence on disconnected consciousness provides compelling support: patients can show neural markers of awareness despite complete behavioral unresponsiveness, illustrating the perils of conflating these levels. The framework helps resolve debates about AI consciousness, animal consciousness, and theoretical integration by making explicit which questions are being addressed. We discuss implications for research practice, funding priorities, and theoretical development, arguing that progress depends partly on clarity about what is being compared.

The Challenge of Comparison

Consciousness research is expanding across an unusually broad range of systems and disciplinary perspectives (Godfrey-Smith, 2016; Dehaene et al., 2017; LeDoux et al., 2020; Birch et al., 2020). While empirical work has advanced considerably, theoretical convergence remains elusive. We suggest that at the core of these theoretical divergences lies a form of *conceptual misalignment*: different research programs often appear to disagree about consciousness when in fact they are addressing different explanatory targets. Some research focuses on how internal mechanisms generate conscious experience, others on how to detect and measure external manifestations of consciousness, and still others cross the boundaries between biological and artificial systems. These differences are not always made explicit, leading to apparent theoretical conflicts that may actually reflect dialogue barriers between different explanatory questions rather than genuine empirical contradictions.

Two dimensions appear particularly important for situating these different explanatory projects: the level of analysis (internal vs. external) and the nature of the system under study (biological vs. artificial/symbolic). Making these dimensions explicit may help clarify what is being compared across different approaches.

Level of Analysis: Internal Mechanisms vs. External Manifestations

Research on consciousness can focus on different levels of analysis. Some approaches investigate the internal mechanisms that may support or constitute conscious processing, such as the neural, computational, or informational structures (Nobre & Gresch, 2025).

Others examine **external observable indicators**, such as behaviors, reports, task performance, and other measurable outputs. These indicators are used to *infer or detect* the presence or characteristics of consciousness, rather than being directly equated with consciousness itself. Importantly, the relationship between these external markers and internal conscious states is itself a theoretical question requiring explanation.

Both perspectives address legitimate scientific questions, but they target different explanatory goals. Mechanism-focused accounts ask how conscious states might be generated or organized within a system. Manifestation-focused accounts examine how such states, if present, might be expressed, reported, or detected. These are complementary rather than competing explanatory projects, though the relationship between them remains an important theoretical question.

The Risk of Category Errors

Serious inferential errors can arise when evidence from one level is inappropriately used to answer questions at another level. A typical example is using the absence of behavioral responsiveness to infer the absence of conscious experience.

However, internal conscious experience and external behavioral responsiveness can **dissociate**. Research on 'disconnected consciousness' during anesthesia demonstrates that subjects may report vivid subjective experiences despite complete absence of behavioral responsiveness during the same period (Sanders et al., 2012; Mashour et al., 2020). These studies reveal that the internal dimension of consciousness (subjective experience) and its external dimension (behavioral output) are not necessarily synchronized.

Similar issues arise in clinical neurology. Patients diagnosed with unresponsive wakefulness syndrome may be completely behaviorally non-responsive yet show neuroimaging evidence of covert awareness. For instance, Owen et al. (2006) and Monti et al. (2010) demonstrated that such patients could modulate brain activity by imagining playing tennis or navigating through rooms, despite being unable to produce any external behavior. Subsequent work has further documented such dissociations using bedside detection methods (Cruse et al., 2011) and identified distinct neural networks mediating awareness of environment versus self (Vanhaudenhuyse et al., 2011; Boly et al., 2008). These findings demonstrate that **the absence of behavioral evidence cannot be directly used to deny the existence of internal conscious states**.

Therefore, researchers must be vigilant against this category error: directly equating evidence (or absence of evidence) at the external manifestation level with conclusions at the internal mechanism level. One core function of this framework is to help identify and avoid such erroneous inferences.

Altered states such as meditation, hypnosis, or lucid dreaming also highlight variations in the relationship between internally generated experience and externally oriented responsiveness (Demertzi et al., 2011), further illustrating how different aspects of consciousness may be emphasized depending on the analytical focus.

System Type: Biological vs. Artificial and Symbolic Substrates

A second dimension concerns the kinds of systems being investigated. Consciousness research has traditionally focused on biological organisms, particularly humans and non-human animals. More recently, discussion has extended to artificial systems and symbolic forms, including machine learning models, linguistic artifacts, and cultural representations (Gamez, 2020; Kirmayer et al., 2020).

These system types differ in potentially relevant ways: in their physical implementation, developmental history, evolutionary background, and relationship to biological needs. Importantly, including artificial systems in this framework does not assume they are conscious; rather, it acknowledges that claims about consciousness are increasingly made with reference to such systems, making it useful to situate these claims within a common analytical space.

The question of which features are necessary or sufficient for consciousness may receive different answers depending on which kinds of systems are considered. Making this dimension explicit helps clarify the scope and intended generality of different theoretical claims.

A Descriptive Framework

These two dimensions (level of analysis and system type) define a simple descriptive space for consciousness research (Figure 1). The horizontal axis distinguishes mechanism-focused from manifestation-focused approaches. The vertical axis differentiates biological from artificial/symbolic systems.

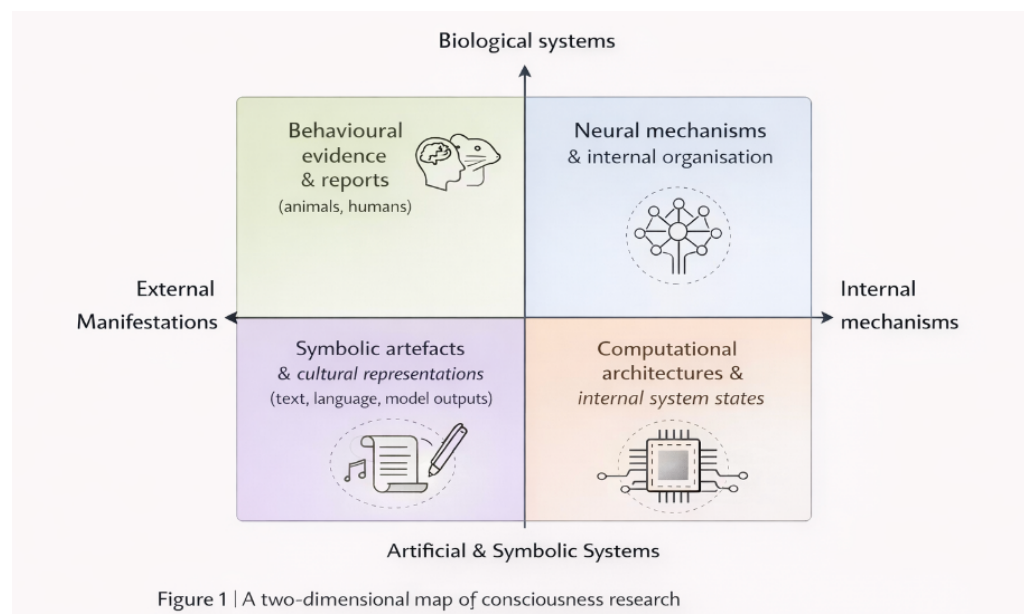


Figure 1. The horizontal axis distinguishes approaches focused on internal mechanisms from those centred on external manifestations such as behaviour. The vertical axis differentiates biological substrates from artificial and symbolic systems. Together, these dimensions define four conceptual quadrants that situate diverse research programmes and claims about consciousness without presupposing a shared definition or theoretical commitment.

This framework comprises four quadrants, each corresponding to different research foci. For example: (1) research on internal mechanisms in biological systems investigates how neuronal assemblies generate integrated information (e.g., Global Workspace Theory or Integrated Information Theory); (2) research on external manifestations in biological systems examines how animals exhibit self-recognition behavior in mirror tests; (3) research on internal mechanisms in artificial systems analyzes whether attention mechanisms in deep learning models possess functional integration; (4) research on external manifestations in artificial systems evaluates the coherence and consistency of AI systems in dialogue. These examples illustrate that different research questions actually occupy different analytical spaces.

Value of the Framework

This framework aims to defuse and reframe several longstanding debates:

- **False opposition between theories:** The debate between Global Workspace Theory (GWT) and Integrated Information Theory (IIT) partially stems from the former focusing more on functional external manifestations (information broadcast and access for reporting and behavioral control), while the latter emphasizes internal information integration structures. These two theories may not be entirely contradictory but rather explain phenomena at different levels of analysis.
- **Clarifying AI consciousness debates:** Debates about whether large language models are conscious often conflate behavioral capacities (e.g., generating coherent text) with internal experiential states. This framework helps distinguish: whether a system exhibits consciousness-like behavior externally versus whether the system possesses subjective experience internally are two different questions.
- **Cross-species comparison dilemmas:** In animal consciousness research, behavioral evidence (e.g., mirror self-recognition) is often used to infer internal conscious states. This framework makes explicit that such inference crosses different analytical levels and requires additional theoretical bridging.

By making these distinctions explicit, researchers can more clearly identify whether disagreements stem from genuine empirical differences or from addressing different explanatory questions.

Clarificatory Functions

This framework serves several clarificatory functions:

1. **Situating research questions:** It helps identify which explanatory project a given claim or theory addresses, without prejudging its validity.
2. **Clarifying disagreements:** Apparent conflicts between views may reflect different quadrants of focus rather than incompatible empirical predictions.
3. **Identifying evidence types:** Different quadrants may call for different forms of evidence and different standards of inference.
4. **Supporting cross-talk:** By making analytical differences explicit, the framework may facilitate more productive exchange between approaches.

The framework does not adjudicate between theories or privilege any particular definition of consciousness. Rather, it offers a tool for clarifying what is being investigated and compared.

Relationship to Existing Meta-Frameworks

This framework is both related to and distinct from other meta-analytical frameworks in cognitive science, particularly Marr's levels of analysis. Marr (1982) distinguished computational, algorithmic/representational, and implementational/hardware levels, all concerning different explanatory levels **within a single system**.

In contrast, our framework distinguishes: (1) **explanatory focus** (mechanism vs. manifestation): whether attention is on how the system operates internally or on observable external outputs; and (2) **system type** (biological vs. artificial): spanning different physical implementation bases across domains.

Thus, Marr's framework answers 'how to explain the same cognitive ability at different abstraction levels,' while our framework answers 'what different research programs are actually studying regarding consciousness.' The two are complementary: Marr's levels can be further applied within each quadrant of our framework.

Another relevant distinction is Ned Block's differentiation between 'access consciousness' and 'phenomenal consciousness' (Block, 1995). Access consciousness refers to functional accessibility of information for reporting and behavioral control, while phenomenal consciousness refers to the subjective 'qualia' of experience. This distinction illustrates that debates in consciousness research often arise when researchers focus on functional accessibility versus subjective experience. Our framework extends this insight at a broader level: not only are there multiple dimensions internally (functional vs. experiential), but internal-external and biological-artificial dimensions also affect how we define and study consciousness.

Illustrative Examples

The following examples demonstrate how this framework can clarify debates by making explicit which analytical spaces different approaches occupy.

Case 1: Global Workspace Theory vs. Integrated Information Theory

Apparent disagreement: These two prominent theories appear to offer different answers about the neural basis of consciousness.

Framework analysis: GWT primarily addresses the **Biological × External** quadrant: how conscious content is broadcast to enable reporting and behavioral control. IIT primarily addresses the **Biological × Internal** quadrant: the intrinsic information integration structure that may constitute conscious experience.

Clarification: These theories may each be valid within their respective explanatory domains. Their apparent conflict partially stems from differences in explanatory target rather than incompatible empirical predictions. This does not resolve substantive disagreements between them, but clarifies that some of the debate reflects different analytical foci.

Case 2: Mirror Self-Recognition in Animals

Apparent disagreement: Can mirror self-recognition tests demonstrate that animals possess self-consciousness?

Framework analysis: Mirror tests provide evidence in the **Biological** × **External** quadrant (behavioral responses to visual self-information). Questions about self-consciousness concern the **Biological** × **Internal** quadrant (subjective experience of self).

Clarification: Inferring internal states from external behavior requires additional theoretical assumptions. Different researchers hold different positions on how strong this inference can be, which explains why identical behavioral data lead to different conclusions. The framework does not resolve this inferential question but makes explicit where the analytical gap exists.

Case 3: Consciousness in Large Language Models

Apparent disagreement: Does the ability of AI systems to generate human-like dialogue indicate consciousness?

Framework analysis: Conversational abilities belong to the **Artificial** × **External** quadrant (observable output patterns). Questions about subjective experience belong to the **Artificial** × **Internal** quadrant.

Clarification: Similarity in external manifestations does not entail equivalence in internal states. The framework helps distinguish: (1) whether AI exhibits functional indicators of consciousness, from (2) whether AI possesses subjective experience. These are distinct questions that may require different forms of evidence and argumentation.

Case 4: Free Energy Principle and Active Inference

Apparent scope: The Free Energy Principle (FEP) and active inference are proposed as unifying frameworks that may span multiple levels of explanation.

Framework analysis: FEP primarily addresses the **Biological** × **Internal** quadrant: how biological systems implement variational inference through internal generative models and prediction error minimization. However, FEP also makes claims in the **Biological** × **External** quadrant: explaining observable behavioral patterns, action selection, and learning through the mathematics of free energy minimization.

Clarification: The framework clarifies that FEP can function as both a theory of internal computational mechanisms (how brains implement Bayesian inference) and a theory explaining external behavioral manifestations (why organisms act adaptively). These are related but distinct explanatory projects. Debates about FEP sometimes conflate claims about internal workings with claims about external behavior, and making this distinction explicit can help situate different interpretations and applications of the principle.

These examples illustrate how the framework helps clarify what appears to be theoretical disagreement but may reflect different analytical foci. By making explicit which quadrants different approaches occupy, researchers can better identify whether disagreements are empirical, conceptual, or reflect different explanatory interests. This clarification does not resolve substantive

debates, but it can prevent misunderstandings that arise from comparing theories addressing different explanatory targets.

When Distinctions Matter: Avoiding Quadrant Conflation

The value of this framework becomes particularly apparent when it helps identify which quadrants are being conflated, thereby clarifying the source of disagreement or confusion.

Human-AI Interaction and Personality Attribution

Users often attribute personality-like consistency to AI systems across interactions. These observations are variously interpreted as evidence of mental states, anthropomorphic projection, or behavioral regularities. **Quadrant conflation:** Conflating **Artificial × External** (observed behavioral consistency in system outputs) with **Artificial × Internal** (whether the system possesses internal mental states analogous to human personality).

Framework clarification: The disagreements often reflect different analytical starting points (mechanism vs. manifestation, biological vs. artificial) rather than different empirical observations. Recognizing this distinction helps identify whether debates concern the interpretation of agreed-upon behavioral patterns or genuine empirical differences.

Cross-Species Consciousness Comparison

Debates about consciousness in animals sometimes conflate questions about internal mechanisms with questions about behavioral indicators. **Quadrant conflation:** Treating evidence from **Biological × External** (behavioral responses, task performance) as if it directly settles questions in **Biological × Internal** (subjective experience, phenomenal states).

Framework clarification: Disagreements persist partly because different forms of evidence are being privileged. Some researchers require stronger theoretical bridging before accepting behavioral evidence as indicative of internal states; others consider the inference more straightforward. Making this analytical gap explicit does not resolve the disagreement but clarifies its nature.

Theoretical Integration Challenges

Attempts to synthesize different theories of consciousness may founder when those theories address different levels of analysis or different system types. **Quadrant conflation:** Expecting theories positioned in different quadrants (e.g., GWT in **Biological × External** and IIT in **Biological × Internal**) to make identical predictions, analogous to demanding that gravitational theory and electromagnetic theory explain the same phenomena in the same way.

Framework clarification: Theories may be compatible within their respective explanatory domains even if they appear incompatible when forced into the same analytical space. Making analytical distinctions explicit does not resolve substantive disagreements, but it can help identify whether disagreements are empirical, conceptual, or reflect different explanatory interests.

Box 1 | Empirical Mapping of the Consciousness Literature

To test whether the proposed framework captures meaningful structure in consciousness research, we systematically mapped 1,336 papers published before 2026 onto the two-dimensional space.

Coding procedure: Two independent raters coded each paper's primary focus along both dimensions using continuous scales (Internal ↔ External; Biological ↔ Artificial), with disagreements resolved through discussion. Papers were then assigned to discrete grid cells based on threshold criteria.

Distribution patterns: The corpus shows uneven coverage across the framework space (Figure Box 2A). The largest concentration falls in the Biological × Ambiguous cell (267 papers, 22.5%), reflecting traditional neuroscience approaches that blur internal mechanisms with external behavioral measures. Pure external-artificial work remains rare (20 papers, 1.7%), while internal-biological research dominates traditional consciousness neuroscience (267 papers combining internal-biological and ambiguous-biological cells).

Continuous semantic structure: Density analysis reveals the corpus forms coherent clusters rather than uniform coverage (Figure Box 2B). High-density regions emerge around: (1) biological internal mechanisms, (2) biological external manifestations, and (3) ambiguous middle-ground approaches. The artificial dimension remains sparsely populated, with recent AI consciousness debates contributing to emerging density.

Interpretation: This distribution suggests the framework captures real structure in how researchers approach consciousness. The concentration in "ambiguous" cells supports our core claim: many apparent theoretical disagreements reflect implicit conflation of internal and external levels rather than explicit methodological choices. The sparse artificial-external corner indicates an emerging research frontier where behavioral AI capabilities increasingly prompt consciousness attributions.

Biological Ambiguous Artificial	16.4%	22.5%	7.3%
	3.2%	15.1%	9.5%
	1.7%	15.0%	9.4%
	External	Ambiguous	Internal

Figure Box 2A. Heatmap showing paper distribution across the 3×3 semantic grid

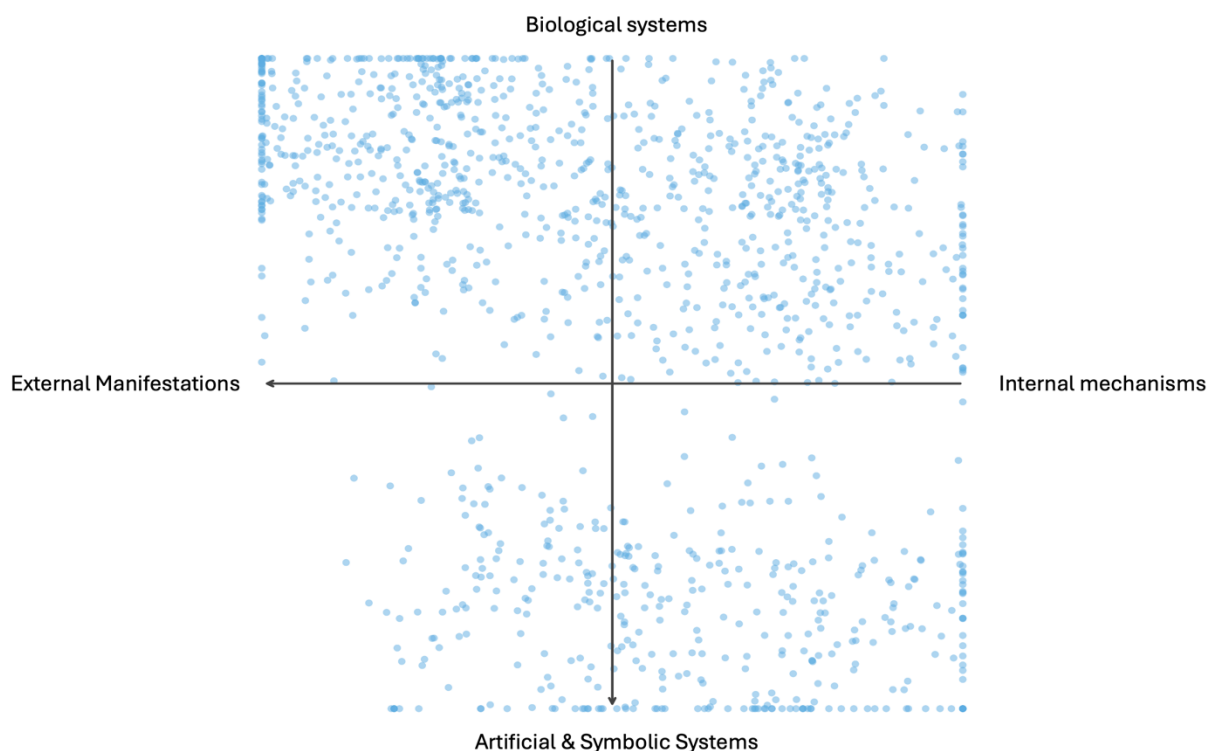


Figure Box 2B | Continuous semantic placement showing density contours of the pre-2026 corpus

Link to main framework: These empirical patterns validate the framework's descriptive utility. The uneven distribution is not arbitrary. It reflects genuine differences in explanatory targets. Making these implicit differences explicit, as the framework does, helps clarify whether research programs genuinely conflict or simply address different quadrants of the consciousness problem.

Box 2 | Temporal Structure Across Analytical Contexts

Time features prominently in consciousness research, but temporal concepts function differently across the analytical framework. Rather than adding a third dimension, temporal structure is realized differently in each quadrant, shaped by the level of analysis and system type:

Biological × Internal: Experienced continuity, autobiographical memory, and the sense of temporal flow inherent to first-person perspective (Gallagher, 2017; Zahavi, 2019). This is lived temporality rather than measured time.

Biological × External: Measurable sequences, i.e., reaction times, learning curves, behavioral regularities. This third-person temporal structure supports experimental investigation and cross-individual comparison (Krakauer et al., 2017).

Artificial × Internal: Time encoded in state transitions, model updates, and training history. This produces temporal structure without constituting experienced duration—a form of time that is neither subjective nor absent, but implemented differently (Clark, 2016; Sutton & Barto, 2018).

Artificial × External: Time emerges through output sequences, turn-taking, and conversational rhythm. Here temporal structure is interpreted through human engagement rather than implemented within the system (Pickering & Garrod, 2021).

Link to main framework: These different realizations of temporal structure are not interchangeable. Transferring temporal concepts across analytical contexts (e.g., applying 'lived temporality' to AI output sequences) constitutes a category error. Recognizing these distinctions helps avoid misunderstandings about the role of time in consciousness research and reinforces the framework's central point: concepts appropriate to one quadrant may not transfer directly to another.

Conclusion

The framework proposed here is offered as a clarificatory tool rather than a theoretical position. By making explicit two key dimensions along which consciousness research varies—level of analysis (mechanism vs. manifestation) and system type (biological vs. artificial/symbolic)—it aims to help identify what different approaches are investigating and where their questions diverge.

As consciousness research continues to diversify in its methods, systems of study, and disciplinary perspectives, progress may depend partly on clarity about what is being compared. The framework does not resolve debates about consciousness, but it may help situate them more productively within a shared analytical space.

This approach is compatible with multiple theoretical positions and does not presuppose answers to substantive questions about the nature or distribution of consciousness. Rather, it provides a descriptive structure for organizing different explanatory projects and the relationships between them. Whether such clarification proves useful will depend on whether it helps researchers identify productive points of contact and meaningful disagreement across different approaches to consciousness.

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